

Cross Evaluation of Capacity Optimization Algorithms on Microgrids

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Background

The objective of the Half Moon Bay microgrid project is to use a 100% electrified renewable energy generation and storage system to meet demands with minimum loss of load. The renewable energy generation and storage system consists of solar PV panels, battery, heat pump, and phase-changing material (PCM) thermal storage. The Berg only consumes electricity to meet heating, cooling, ventilation, and other plug and miscellaneous loads.

In order to achieve minimum loss of load during system operation while minimizing the levelized cost of the system, it is necessary to use optimization to size the system capacities including solar PV, battery, and thermal storage units. With any capacity optimization model, the optimal capacities are heavily dependent on the input data, which in this case, the weather profiles. Therefore, it is necessary to choose a capacity optimization model that considers the variability in the weather profiles.

Literature Review

There are studies focusing on using a specific stochastic or robust capacity optimization model to size a microgrid or multi-microgrid, but the cross evaluation of different capacity optimization algorithms is unique. Also, most of the microgrids in literature are not islanded and therefore have the option to buy and sell electricity, the Half Moon Bay microgrid is completely islanded, therefore, the focus is not to oversize the generation capacity and sell electricity, but to minimize the loss of load.

Currently, most studies employ two-stage or multi-stage stochastic optimization models, utilizing various simulation algorithms to generate synthetic scenarios for uncertainty modeling (environmental factors, demand variation, etc.). The modeling of uncertainties is a crucial component in these algorithms, yet few studies provide compelling reasoning for their specific design choices. This highlights the need for a thorough comparison of these approaches to better understand their effectiveness and underlying assumptions.

Methodology

In order to account for the data variability, the dataset will be divided into a training dataset and a testing dataset, each containing multiple yearly weather profiles from NSRDB at the microgrid site. There will be in total 5 different capacity optimization algorithms, and each of the 5 capacity algorithms will be used to optimize the optimal capacities of the microgrid system based on overall cost (levelized cost of capacity + penalty for loss of load) using the training dataset. Then, each of the resulting set of system capacities will be used to simulate through the testing

dataset, and the overall cost (levelized cost of capacity + penalty for loss of load) of the testing dataset will be the final performance score for each capacity optimization algorithm.

Data

Training data: the training dataset consists of 20 years of NSRDB Half Moon Bay weather data. These 20 years are randomly drawn from the year 1998 to 2022. The training dataset can be later manipulated in the following ways if suitable for the capacity optimization algorithm:

1. Yearly continuous weather data (nothing gets done to the original training data)
2. Broken up into segments of certain horizon (weeks, months) and rearranged
3. Used as a base profile to generate more scenarios based on probability distributions.
4. Used as a base profile to generate more scenarios following stochastic regression models

Testing data: the remaining 5 years of the NSRDB Half Moon Bay weather data between 1998 and 2022 are taken as the testing dataset. The capacity optimization algorithms have not seen these 5 years of weather data, but the resulting capacity will be simulated through these 5 years of data.

Capacity Optimization Algorithms

These are the potential capacity optimization algorithms that this paper will aim to test their performances on different data scenarios.

1. Perfect forecast + simple LP
 - a. The benefit of simple LP is that it is really easy to implement and the computation speed is the fastest.
 - b. The disadvantage of simple LP is that there is no robustness in the resulting system capacity.
2. Probability distribution on weather + Two-stage stochastic optimization

With two-stage stochastic optimization, we minimize expected costs across a discrete set of scenarios, which could be sampled from distribution or based on historical data.

 - a. The benefit of stochastic optimization is that it takes into account a wide range of data, which minimizes the expected costs.
 - b. The disadvantage of stochastic optimization is that runtime will be very long.
3. Robust optimization

With robust optimization, we optimize for the worst case scenario and find a set of solutions that meets constraints for all outcomes of uncertainty.

 - a. The benefit of robust optimization is that the resulting system capacity will have the minimal loss of load.
 - b. The disadvantage of robust optimization is that the system will likely be oversized since we are considering the worst case scenario.
4. Chance constrained-programming

This method involves formulating the optimization problem such that constraints are met with a specified probability (the actual implementation of the algorithm deals with realized instances).

- a. It balances between robustness and cost by allowing a controlled level of risk, avoiding excessive conservatism.
 - b. The complexity of solving chance-constrained problems can be high, and it requires careful selection of risk levels and probability distributions.
5. Distributionally-robust optimization
This approach optimizes system capacities by considering the worst-case distribution within an ambiguity set of probability distributions. It combines elements of both robust optimization and stochastic optimization.
 - a. Offers a more balanced approach by protecting against worst-case distributions while not being overly conservative.
 - b. The computational effort required is significant, as it involves optimizing over a set of possible distributions rather than a single distribution or set of scenarios.
 - c. This algorithm is similar to algorithm 3, so only one of them might be sufficient for the paper.

Objective

Minimize (levelized cost of the system + penalty for loss of load)

Decision Variables

System capacity (size of PV, battery, PCM hot and cold storages) + other time series decision variables

Constraints

Energy conservation constraints, power balance constraints ...

Results

Show the performance score of each capacity optimization algorithm by simulating their respective system capacity (trained from the training dataset) through the test dataset. We could do this for a few different times and take average (we re-choose the 20 years of training data from 25 years each time)